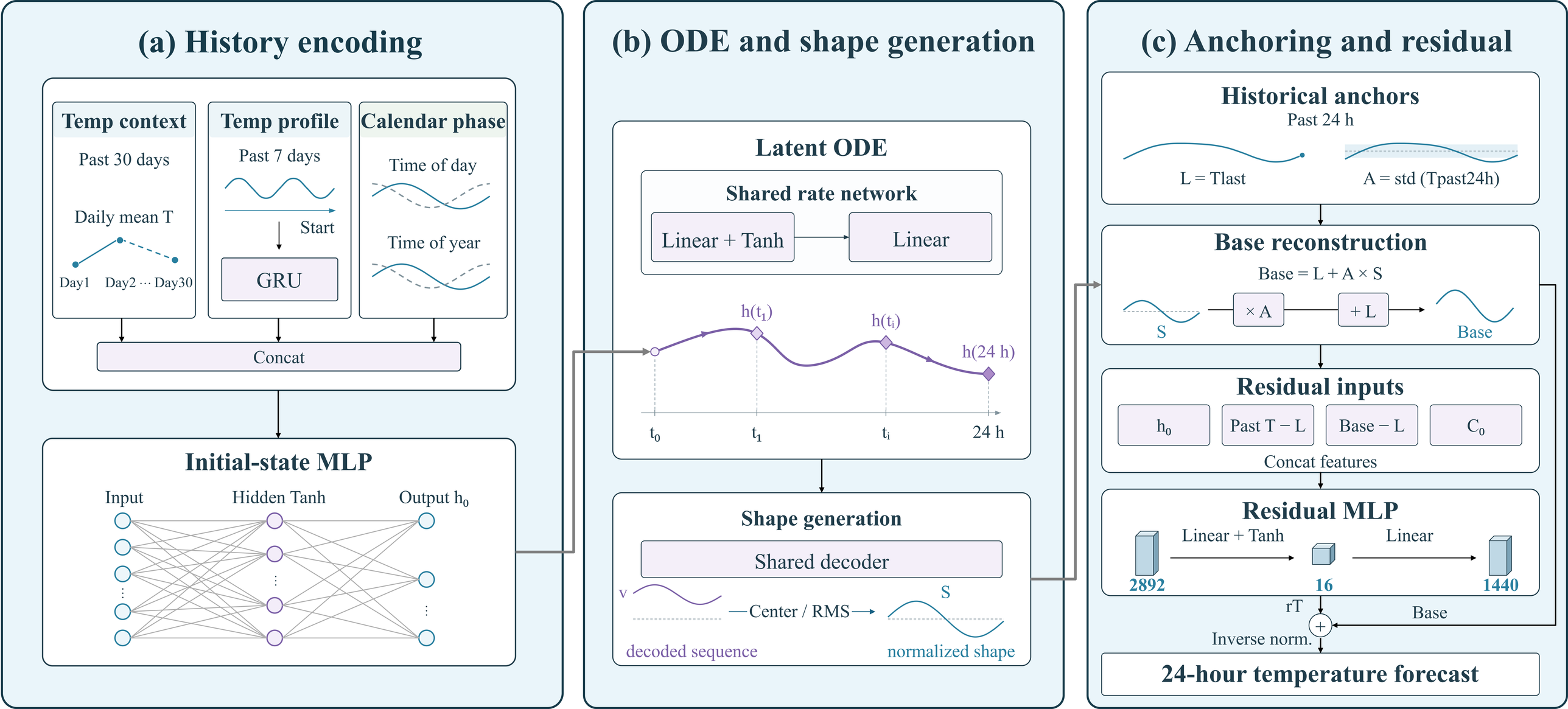

(a) History encoding
Temp context
Past 30 days
Daily mean T
Day1
Day2 ··· Day30
Temp profile
Past 7 days
Start
GRU
Calendar phase
Time of day
Time of year
Concat
Initial-state MLP
Input
Hidden Tanh
Output h0
(b) ODE and shape generation
Latent ODE
Shared rate network
Linear + Tanh
Linear
h(t1)
h(ti)
h(24 h)
t0
t1
ti
24 h
Shape generation
Shared decoder
v
Center / RMS
S
decoded sequence
normalized shape
(c) Anchoring and residual
Historical anchors
Past 24 h
L = Tlast
A = std (Tpast24h)
Base reconstruction
Base = L + A × S
S
× A
+ L
Base
Residual inputs
h0
Past T − L
Base − L
C0
Concat features
Residual MLP
Linear + Tanh
Linear
2892
16
1440
rT
Base
Inverse norm.
24-hour temperature forecast